A Mai Ly



Senior full-stack & backend engineer

I build products end to end for European teams, with most of my depth in the backend: APIs, realtime systems and AI features that stay up in production.

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About

Since 2020 I've been an engineer at Digital Unicorn, a French software studio. I've worked on more than 20 of its products, usually as the main backend engineer: NestJS and Node APIs, payment flows, realtime features, and LLM and voice integrations, deployed on AWS and Azure with Docker and Kubernetes.

Experience

2024 - now

Senior full-stack engineer

Digital Unicorn Co., Ltd, full-time

- Lead engineer on Nutrissimo, a realtime multiplayer game for Institut Pasteur de Lille that launches inside Moodle through LTI 1.3.
- Kept NightPass ticketing up during live events as backend lead: fixed a Redis out-of-memory incident and made rolling deploys safe.
- Built AI products end to end: Days Legacy, RP Tech AI, SparkPlan with its voice agents, and CLS AI.
- Own architecture, delivery and production reliability on the projects I lead.

2020 - 2024

Backend and full-stack engineer

Digital Unicorn Co., Ltd, full-time

- Main backend engineer on WeShake (Stripe and MangoPay payments) and OOVOOM (ran the AWS EKS and Helm deployments).
- Lead backend engineer on BeParentalis, a parenting app with about 20k downloads, and on RP Tech AI.
- Also built DeliverLIB, Just Meat, Raisy, Kibo and CyLimit: payments, realtime features, token vesting and role-based access.

2019 - 2020

Software developer

Phuc Quan Service Trading Co., Ltd

Designed relational databases and built Node.js REST APIs. Researched tools, tuned performance, and debugged and refactored existing modules.

2019

Software developer

G-Lance

My first job. Built Nail Schedule, a booking product (June to December 2019): Node.js REST APIs and the database schema.

Selected work

Nutrissimo

2026 - now

Lead full-stack engineer, for Institut Pasteur de Lille

- Built the server-authoritative game engine, so a class keeps playing through dropped connections: presence, reconnection and idempotent state updates.
- Implemented LTI 1.3 single sign-on so the game launches straight from a Moodle course, with no separate account.
- Work across the whole Turborepo: the Flutter app, the Next.js admin and Supabase.

Flutter, Next.js 15, Supabase Realtime, Turborepo, LTI 1.3, Cloudflare

NightPass

nightpass.fr

Backend lead, ticketing for venues and events. About 10k users on web and mobile.

- Stopped a production out-of-memory crash: traced it to a Redis reconnect storm, fixed the config and added leader locks so rolling deploys no longer drop traffic.
- Kept entry fast with many doors scanning at once by reworking the QR scan path: back-pressure, compare-and-set writes and readiness probes.

NestJS, GraphQL, Prisma, PostgreSQL, Redis, BullMQ, Stripe, AWS ECS

SparkPlan

sparkplan.ai sparkplanvoice.com

AI engineer, budget advisor running on Sortlist.com. About 5k reports generated.

- Built the OpenAI, LangChain and ChromaDB pipeline that turns a brief into an estimate, routing between GPT-4o, GPT-4 and GPT-3.5 and answering in five EU languages.
- Built the voice agents that qualify leads by phone, from telephony and WebSocket streaming to the emailed report.

NestJS, OpenAI, Azure OpenAI, LangChain, ChromaDB, WebSocket, MongoDB, Stripe

RP Tech AI

Backend architect, AI productivity suite for French users. About 8k users.

- Put four AI providers behind one API: OpenAI, Rev for speech-to-text, Ordalie for legal answers and DeepL for translation, with LangChain and ChromaDB for retrieval.
- Built paid tiers with usage tracking on Stripe, and the API behind the browser extension.

NestJS, OpenAI, Rev, DeepL, Ordalie, LangChain, ChromaDB, MongoDB, Stripe

WeShake

weshake.io

Senior backend engineer, peer-to-peer payments and cards. About 12k users.

- Moved money through two processors, Stripe and MangoPay: transfers, payment links, recurring billing and webhooks.
- Shipped GDPR data deletion, multi-device login with OTP, and the referral and bonus system.

Node.js, Express, MySQL, Sequelize, Stripe, MangoPay, Socket.IO, Twilio, AWS, Kubernetes

Other projects

CLS AI for Promecano
Voice and chat assistant that diagnoses industrial machine errors from a knowledge base for each device.

BeParentalis
Parenting app reviewed by pediatricians, about 20k downloads. Lead backend: data model, Algolia search, RevenueCat subscriptions.

OOVOOM
ERP for 100+ partner garages. Role-based access, automated quotes and invoices, AWS EKS and Helm deployments.

Days Legacy
AI advisor that combines a user's portfolio with live financial news to answer wealth and legacy planning questions.

Eddwin
Training course marketplace: multilingual Next.js front end, onboarding, email automation and Stripe billing.

CyLimit
Fantasy cycling with rider NFTs. I built the live announcements, notifications and media pipeline.

Skills

Core stack	**TypeScript, Node.js, NestJS, React, Next.js, Flutter, PostgreSQL, MongoDB, Redis, GraphQL, Docker, Kubernetes, AWS**
Backend	REST and GraphQL APIs, WebSockets and SSE, microservices with queues (BullMQ, Kafka), OAuth2 and RBAC, Stripe and MangoPay payments.
AI and voice	OpenAI and Azure OpenAI, LangChain, RAG with ChromaDB, Retell AI and ElevenLabs voice agents.
Cloud and DevOps	AWS (ECS, EKS, Lambda, S3), Azure AKS, Helm, ArgoCD, GitLab CI, Cloudflare.

Education & certificates

Top Notch 2 English certificate, CEFR level B2

FPT Polytechnic, Software applications, 2016 – 2018. GPA 7.8/10.

Vietnamese: Native **English:** Professional working proficiency (B2)